Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2025

Course: B.Tech | *Time: 03 Hours*
Code: 104502 | *Subject: Digital Signal Processing* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Choose the correct answer the following (Any seven question only):** **[2 x 7 = 14]**

(a) The output of an LTI system can be determined using
(i) Sampling (iii) Quantization
(ii) Convolution (iv) Modulation

(b) Which of the following is an advantage of DSP over analog systems?
(i) More noise (iii) Better accuracy and reliability
(ii) Less flexibility (iv) Difficult storage

(c) A system is said to be LTI if it is
(i) Linear and Time Invariant (iii) Logical and Time Invariant
(ii) Linear and Time Variant (iv) Linear and Tuned

(d) Convolution is used to determine
(i) Frequency response (iii) Sampling frequency
(ii) System output (iv) Fourier coefficients

(e) The DTFT of a discrete-time signal is
(i) Always periodic (iii) Finite
(ii) Non-periodic (iv) Real only

(f) Aliasing occurs when the sampling frequency is
(i) Greater than Nyquist rate (iii) Less than Nyquist rate
(ii) Equal to Nyquist rate (iv) Double the signal frequency

(g) FFT is mainly used to
(i) Increase noise (iii) Amplify signals
(ii) Reduce computational complexity of DFT (iv) Sample analog signals

(h) Correlation is used to measure
(i) Power consumption (iii) Frequency distortion
(ii) Similarity between signals (iv) Sampling error

(i) The DFT converts a signal from
(i) Time domain to frequency domain (iii) Analog to digital
(ii) Frequency domain to time domain (iv) Continuous to discrete amplitude

(j) Nyquist sampling theorem states that sampling frequency should be
(i) Equal to signal frequency (iii) Half the maximum frequency
(ii) Twice the maximum frequency (iv) Four times the signal frequency

Q.2 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain the basic elements of a Digital Signal Processing system with block diagram.

(b) Discuss the advantages of DSP over analog signal processing systems.

Q.3 **Solve both questions :** **[7 x 2 = 14]**

(a) Classify signals into continuous-time and discrete-time signals with examples.

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(b) Define Linear Time Invariant (LTI) systems and explain their properties.

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain convolution sum and determine the response of an LTI system. Use:
$y[n] = x[n] * h[n] = \Sigma_{k=-\infty}^{\infty} x[k]h[n-k]$

(b) Describe discrete-time systems using difference equations.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Define Discrete Time Fourier Transform.

(b) Explain the properties of DTFT with examples. Use:
$X(e^{j\omega}) = \Sigma_{n=-\infty}^{\infty} x[n]\, e^{-j\omega n}$

Q.6 **Solve both questions :** **[7 x 2 = 14]**

(a) Explain frequency domain representation of LTI systems.

(b) Derive the mathematical expression of DFT. Use:
$X[k] = \Sigma_{n=0}^{N-1} x[n]\, e^{-j(2\pi/N)kn}$

Q.7 **Solve both questions :** **[7 x 2 = 14]**

(a) A voice signal contains frequencies up to 4 kHz. Determine the Nyquist sampling rate.

(b) Discuss radix-2 Decimation in Time FFT algorithm.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Compare DFT and FFT with respect to computational efficiency.

(b) Find the 4-point DFT of the sequence:
$x[n] = \{1,0,1,0\}$

Q.9 **Write short notes on any two of the following:** **[7 x 2 = 14]**

(a) FIR filter design

(b) IIR filter design

(c) Discrete time sinusoidal signals.

(d) Frequency domain representation of LTI Systems

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